Moves on the Space of Bilinear Schemes: a Field Guide to Rank-Hunting Search

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Reproducible methods note — logic repo, matmul track, 2026-07-14. Companion to the flower, adds, 53-schemes, and zkML papers; all engines cited live in this repository.

1 The space

Fix a format $\langle m, m, m\rangle$ and a coefficient field K. A **scheme of rank** r is a decomposition of the matrix-multiplication tensor into r rank-one terms,

$$T_m = \sum_{t=1}^{r} a_t \otimes b_t \otimes c_t, \qquad a_t, b_t, c_t \in K^{m\times m},$$

verified by the m^6 Brent equations. The **space** is the union of rank strata: at 3×3 the stratum $r = 27$ contains the naive scheme, $r = 23$ the records (17,376+ known classes over $\mathbb{Z}$), $r = 22$ nothing known; at 4×4 the strata run 64 (naive), 49 (Strassen^2), 48 (dyadic records over odd characteristic), 47 (characteristic 2 only). Two schemes are **equivalent** (de Groote) if a sandwich (PaQ^{-1}, QbR^{-1}, $R\tilde{c}P^{-1}$), a cyclic slot rotation, or a per-summand rescaling maps one to the other; searches operate on gauge-canonical representatives (monic slots over $\mathbb{F}_p$; primitive sign-normalized over $\mathbb{Q}$).

What is known about the *shape* of the space, from the literature and from this repository's certificates:

- **It is disconnected at bounded rank.** Kauers–Moosbauer's walks from the 3×3 naive scheme land in 584 distinct connected components of the rank-≤23 flip graph (64,061 vertices) [1]. Our 4×4 fringe certificate is a completeness proof of confinement: the rank-48 seed's component under solved dyadic moves at rank ≤ 50 is *finite* — 920 states in 16 clusters hanging off the seed as a cut vertex, exhaustively enumerated and closed.
- **Altitude connects, but slowly.** Splits (rank+1 excursions) merge components: our chase explored eight levels above the 4×4 seed (1.45B states) without meeting a second rank-48 class.
- **The field changes the geometry.** Over $\mathbb{F}_2$ the factor value space has 2^{m^2} points, so equal factors and coplanar triples arise by birthday collision as a walk churns; over a 64-bit proof field the value space is effectively infinite, collisions never occur, and only *algebraically solved* moves create the coincidences that descents spend (measured: 448M flips with random and small-dyadic λ from naive over Goldilocks produced zero reductions; the same protocol over $\mathbb{F}_2$ produced ∼60 within minutes).

2 The moves

Notation for the toy examples: e_{ij} is the matrix unit; the naive 2×2 scheme has eight summands $e_{ik} \otimes e_{kj} \otimes e_{ij}$; Strassen's seven summands are labeled $M_1..M_7$ ($M_1 = (e_{11}+e_{22}) \otimes (e_{11}+e_{22}) \otimes (e_{11}+e_{22})$, $M_2 = (e_{21}+e_{22}) \otimes e_{11} \otimes (e_{21}-e_{22})$, $M_6 = (e_{21}-e_{11}) \otimes (e_{11}+e_{12}) \otimes e_{22}$, ...).

M-1. Flip (rank-preserving; the K–M edge). Two summands whose slot-s factors are *equal* (post-gauge) exchange material in the other slots: with shared a and transfer λ,

$$u \otimes b_1 \otimes c_1 + u \otimes b_2 \otimes c_2 \;\to\; u \otimes (b_1+\lambda b_2) \otimes c_1 + u \otimes b_2 \otimes (c_2-\lambda c_1).$$

Toy (naive 2×2, summands (1, 1, 1) and (1, 1, 2), both with $a = e_{11}$, $\lambda = 1$): $b_1 \leftarrow e_{11} + e_{12}$ and $c_2 \leftarrow e_{12} - e_{11}$; the two cross terms cancel and the sum is unchanged. Over $\mathbb{F}_2$, $\lambda = 1$ is the only choice; over $\mathbb{Q}$ we restrict λ to dyadics; over $\mathbb{F}_p$ any $\lambda \ne 0$ is legal.

M-2. Reduction (rank −1, and the characteristic-2 −2). Two summands sharing *two* slots merge: $u \otimes v \otimes c_1 + u \otimes v \otimes c_2 = u \otimes v \otimes (c_1{+}c_2)$; if $c_1 + c_2 = 0$ both vanish — over $\mathbb{F}_2$ this fires whenever two summands become fully identical, a rank −2 event with no analogue over large fields. The $\mathbb{Q}/\mathbb{F}_p$ variant also merges shared-slot pairs whose third slots are *proportional*. Reductions are the only rank-lowering moves.

M-3. Split (rank +1). Rewrite $f_i = \mu f_k + (f_i - \mu f_k)$ in one slot, producing a summand that *shares* that slot with summand k. Toy (μ=1, slot a, M_1 against M_2): $a(M_1) = e_{11} + e_{22} \to$ parts $(e_{21}{+}e_{22})$ and $(e_{11}{-}e_{21})$, each tensored with $b(M_1) \otimes c(M_1)$. Splits buy altitude and manufacture shared pairs (the flip fuel). A fresh split's two parts share *both* other slots, so a reduce-on-sight walk must be gated (the split/merge livelock found building pursue9: reductions legal only after an intervening flip).

M-4. Solved flip (coincidence-targeted; ours). For a shared pair (i, j) and a third summand m, solve $f_i(t) + \lambda f_j(t) = \mu f_m(t)$ by 2×2 Cramer in the transfer slot t: the flip with *that* λ makes summand i's t-factor proportional to m's — an alignment created by algebra, not luck. Toy: $T_1 = u \otimes b_1 \otimes c_1$, $T_2 = u \otimes b_2 \otimes c_2$, $T_3 = u' \otimes (b_1{+}b_2) \otimes c_3$: the pair (T_1, T_2) shares slot a; solving in slot b gives $\lambda = 1$, and after the flip T_1 and T_3 share slot b exactly. Over $\mathbb{F}_2$ solved and random flips coincide in *value* ($\lambda = 1$) and differ only in *selection*; over $\mathbb{F}_p$ the solved λ is one specific field element sampling would never hit — the only reliable coincidence factory (measured, §1).

M-5. Closing move (second-alignment factory; ours). Given an *already aligned* pair, search for a single flip against any third summand that makes a second slot proportional, after which M-2 fires. Over $\mathbb{F}_p$ it needs coplanar triples in the transfer slot — generic states lack them (the structural reason naive-descent fails over big fields); over structured scheme neighborhoods it is routine (every storm quench).

M-6. Sandwich (symmetry) move (planned). Apply $(P, Q, R) \in \mathrm{GL}(m, K)^3$: a gauge teleport. Flips are equivariant, so flip-graph components come in isomorphic families (K–M observe isomorphic components); the sandwich move merges each family into one search component. Cheap and strictly connectivity-increasing.

M-7. Closure jump (planned for $\mathbb{F}_p$; exists over $\mathbb{F}_2$ in `anf`). Fix two of the three factor tensors; the Brent system becomes m^2 independent linear systems in the third — solve exactly. When the current products are linearly independent the solution is unique (no move); when dependent, the solution space is positive-dimensional and the jump teleports across it. Toy: at rank 8 one split above Strassen, the eight products span a 7-dimensional space, so fixing α, β leaves a 1-parameter γ-family — the closure walks it in one step. The strongest field-generic non-local move we know (pure linear algebra).

M-8. Repair hop (built: pursue8 `--repair`). Delete k summands, beam-rebuild the residual in $\leq k - 1$ (record event) or k (lateral hop). Exhaustive over $\binom{r}{k}$ at small k — how the local rigidity certificates were bought (3×3: all 30 census seeds $k \leq 9$ rigid over two fields; 4×4: DPS-48 $k \leq 7$ rigid over Goldilocks, 73.6M subsets). As a walk move ($k = 2$ lateral) it hops between flip components at bounded cost.

M-9. Constructor move (AlphaTensor's game; built: pursue8). Hold a residual tensor, subtract arbitrary rank-one terms until zero. Complete in principle — every scheme reachable — but unguided beam search stalls at the Strassen hump (3×3: terminal count 27; measured), precisely the gap a learned policy filled for DeepMind [4].

M-10. Completion jump (HKS method 2; built: `walk.py` over $\mathbb{F}_2$). Freeze a random subset of a known scheme's bits, re-solve the rest with SLS. Non-local and powerful (our 53-scheme discovery; the ≈715-scheme method-2 basin); field-restricted to where a solver exists — over $\mathbb{F}_p$ the analogue is M-7's linear closure.

3 Search techniques tried (with verdicts)

technique	engine	verdict (measured)
random-walk storms	flip23/48/23p	81 new $\mathbb{Z}$-classes; 63 field-novel patterns
exhaustive closure (BFS)	`--native/--lams`	exact component maps
depth-2 exhaustive fringe	pursue5	840 parents closed, 0 reductions
beam on nearmiss	pursue6 chase	8 levels, 1.45B states, 57× null, 0 conversions
mix-and-quench	pursue7	∼31M walks over 3 fields; no records
constructor beam	pursue8	Strassen-hump stall at 27
exhaustive k-repair	pursue8 `--repair`	rigidity radii 9 / 7 / 6
persistent monotone walk	pursue9	$\mathbb{F}_2$: 64 → 53 then starvation; $\mathbb{F}_p$ sterile
frontier pool ladder	pursue10	$\mathbb{F}_2$: 64 → 49 in 3 min; running
SAT completion	`walk.py+anf`	53 schemes in 6 min; basin ≈715
native-ANF SLS	`anf`	8/10 HKS challenge-1
census-guided seeding	SHORTLIST	mobility metrics; band calibration
null-controlled method	`nmrand`	gradients real (57×), not rank inflation

Planned next: M-6 sandwich moves, $\mathbb{F}_2$ solved-flip targeting, M-7 closure jumps over $\mathbb{F}_p$, M-8 as walk move, Moosbauer–Poole symmetric-subspace walks [2], commutative flipC, learned guidance (the neural track's gmi/gmi_mcgs engines).

4 Classical search algorithms on this space

The rank hunt is single-agent search: **state** = gauge-canonical scheme; **actions** = the moves of §2; **goal** = any state of rank below the record (verification $O(m^6)$). The graph has heavy transpositions (hash canonical forms). Two cost models matter: *hop count* (every move 1) and *altitude* (splits 1, flips 0, reductions −).

BFS. Complete, optimal in hops, memory O(frontier). This *is* our `--native` closure and the fringe/flower enumerations — feasible when the reachable set is 10^6-ish. Use for certificates, not hunting.

DFS / iterative deepening. O(depth) memory; our random walks are memoryless randomized DFS; principled depth-capped DFS appears in cfloor's IDDFS (the adds bounds), where it is exact. For hunting, branching (10^2–10^4) dwarfs useful depth (10^5+): unattractive.

Uniform-cost / Dijkstra / SPF. Unit costs: it is BFS. Under the altitude model (splits 1, flips ε) UCS computes the *minimum-altitude descent* — "how high must any path climb to leave this basin" — the quantity the flower's cut-vertex anatomy probes empirically. Frontier memory confines it to component scale, where it would upgrade "no crossing found at +8" to "no crossing exists below +9."

A*. UCS plus admissible h. The obstruction is honest: an admissible h must lower-bound *path* distance to a smaller-rank scheme, and our rigorous bounds (flattenings, counting floors) bound the *tensor*, not the path. Practice therefore uses the inadmissible cousins: the chase is **greedy best-first** on nearmiss (h only), null-certified to carry signal (57×) yet Goodhart-prone (zero conversions). A concrete admissible-within-radius candidate now exists: h built from the repair-ladder certificates (no scheme within replacement-distance k has smaller rank ⇒ any descent needs $> k$ replacements).

Minimax and α/β. No adversary: minimax proper does not apply. But α/β's soul — cut a branch when a bound proves it cannot beat the incumbent — is **branch-and-bound**, and we already run it: the adds floors prune classes ("≥ 56, stop"), the repair ladders prune subset-trees by projected completion size. The transferable idea: give the rank hunt the same treatment via certified component invariants.

MCTS / MCGS. Best fit, with precedent: AlphaTensor *is* MCTS with a learned prior on M-9's space. Formulation: node = scheme (graph search with transposition table — MCGS); rollout = storm/pursue10 walk; reward = −(best rank reached); selection = UCT. Unguided MCTS must be compared at equal budget against best-of-K walks (our standing methodology note). The repository already carries MCGS scaffolding from the neural track (`gmi`, `gmi_mcgs`); the marginal build is the environment adapter, and the natural first target is the $\mathbb{F}_2$ 47 ladder, where ground truth exists and rollouts are cheap.

algorithm	fits?	as what	status
BFS	yes (bounded sets)	closures, certificates	in use
DFS/IDDFS	partially	cfloor sweeps; walks	in use
UCS/Dijkstra/SPF	yes (altitude cost)	min-altitude escape	possible
A*	blocked on admissible h	greedy best-first today	partial
weighted A*	yes	nearmiss + repair-radius h	planned
minimax	no adversary	—	n/a
α/β spirit (B&B)	yes	floors/repair pruning	in use
MCTS/MCGS	yes	guided constructor/ladder	scaffolding exists

5 One formulation to hold them all

Define the **layered scheme multigraph** $G(K, m)$: vertices are gauge-canonical schemes stratified by rank; intra-layer edges are flips (M-1, M-4, M-6, M-7); downward edges are reductions (M-2); upward edges are splits (M-3); non-local chords are repair and completion hops (M-8, M-10). Every technique of §3 is a traversal policy on G; every algorithm of §4 is a standard policy under a cost model from {hops, altitude, compute}; every certificate we publish is an exactly-explored finite subgraph. The record hunt is: *find any vertex in a layer below the known minimum* — and §1's disconnection results say the decisive question is not which traversal is cleverest but which *edge set* makes the target layer reachable. That is why the move inventory of §2, not the algorithm zoo of §4, is this note's center of gravity.

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